

Rates and Unit Rates

Flagship

Lesson 4-1-flagship

Name: _____

Date: _____

Class: _____

ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players – Booth A charges \$3 for 5 games, Booth B charges \$5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

<i>\$12 for 4 games is a rate</i> Rate A ratio comparing two amounts with different units, like miles per hour.	<i>\$3 per 1 game</i> Unit Rate A rate for just 1 of something, like cost for 1 item.
<i>60 miles per hour means 60 miles for every 1 hour</i> Per For each one. Example: 5 dollars per book.	<i>5 to 3 or 5:3</i> Ratio A way to compare two amounts.
<i>Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil – Pack B is the better buy</i> Better Buy The choice that costs less for each item.	<i>$\\$12 \div 4 \text{ games} = \\3 per game</i> Divide Splitting a total into equal groups.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find a unit rate to compare prices and decide the better buy.

1 Notice

You're comparing ticket prices at different game booths in the arcade.

2 Key idea

I can find a unit rate to compare prices and decide the better buy.

3 Words I'll use

Rate

Unit Rate

Per

Ratio

Better Buy

Divide



Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Booth A charges \$3 for 5 games and Booth B charges \$5 for 8 games. Can you tell which is the better deal just by looking at the signs? Why or why not?

Level 1 support

Start here: You can't compare fairly until you find the cost per 1 game (the unit rate) for each booth.

 **Need a hint?**

Sentence starters (optional)

You ____ tell just by looking because the number of ____ is different.

____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ game.

Para comparar de forma justa necesito el costo por ____ juego.

Word bank: rate unit rate per better buy compare

Listen for: Students recognize the booths offer different numbers of games, so a fair comparison needs the cost per 1 game (unit rate).

Level 2

Some players pick Booth B because \$5 buys more games. Why is that reasoning not enough to know the better deal? Justify using cost per game.

- Buying more games does not always mean a better deal because ____.
- The real test is the ____, since ____.

EXPLORE

How did you find the cost per game for each booth, and which booth wins?

Level 1 support

Start here: Divide the price by the number of games to get the unit rate, then compare the two rates.

 **Need a hint?**

Sentence starters (optional)

Booth A costs ____ per game because 3 divided by 5 equals ____.

El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

The better buy is Booth ____ because its ____ is lower.

La mejor compra es el puesto ____ porque su ____ es más baja.

Word bank: **rate** **unit rate** **per** **better buy** **divide**

Listen for: Students compute \$0.60/game for Booth A and \$0.625/game for Booth B, then choose Booth A because the lower unit rate is the better buy.

Level 2

If a player only wants 5 games total, does the better unit rate always give the cheaper total? Explain when it might not.

- The lower unit rate gives the cheaper total only when ____.
- It might not be cheaper if ____.

CONNECT

At the snack bar, how would you use a unit rate to find the best popcorn deal among different sizes?

Level 1 support

Start here: Find the price per ounce (or per cup) for each size, then pick the lowest unit rate.



Need a hint?

Sentence starters (optional)

I would find the price per ____ for each popcorn size.

Hallaría el precio por ____ para cada tamaño de palomitas.

The best deal is the one with the ____ unit rate.

La mejor oferta es la que tiene la ____ tasa unitaria.

Word bank:

rate

unit rate

per

better buy

compare

Listen for: Students describe dividing price by size to get a per-unit cost and choosing the lowest unit rate as the best deal.

Level 2

Why might the biggest popcorn not always be the best value? Use the idea of unit rate to explain.

- The biggest size is not always the best value because ____.
- The unit rate shows the real value because ____.

Guided Examples

Example 1

A booth charges \$4.50 for 9 games. What is the unit rate?

Solution: $\$4.50 \div 9 = \0.50 per game.

Answer: A. \$0.50 per game

Example 2

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

Solution: $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon.}$

Answer: A. 30 miles per gallon

Example 3

A pack of 8 markers costs \$6.00. What is the cost per marker?

Solution: $\$6.00 \div 8 = \0.75 per marker.

Answer: A. \$0.75

Write About the Math

The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

1. Kernel Sentence subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.

Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.

Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rate matters in math

Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rate matters in math because ____.

Tasa importa en matemáticas porque ____.

but
pero

Rate matters in math, but ____.

Tasa importa en matemáticas, pero ____.

so
entonces

Rate matters in math, so ____.

Tasa importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question
Pregunta

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

You ____ **tell just by looking because the number of** ____ **is different.**

____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ **game.**

Para comparar de forma justa necesito el costo por ____ juego.

Booth A costs ____ **per game because 3 divided by 5 equals** ____.

El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which token pack from the table above is the best deal?

- A. Pack A
- B. Pack B
- C. Pack C
- D. They are all the same

Show your work:

2. Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

- A. Runner B
- B. Runner A
- C. Same rate
- D. Cannot determine

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of \$____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** B. Pack B — *Pack B has a unit rate of \$0.20 per token, which is the lowest cost per token.*
2. **Try It 2:** A. Runner B — *Runner A: $4 \div 12 \approx 0.33$ laps/min. Runner B: $5 \div 14 \approx 0.36$ laps/min. Runner B runs slightly more laps per minute.*
3. **Exit Ticket:** A. 6 for \$1.50 — \$0.25 each — $\$1.50 \div 6 = \0.25 per pencil. $\$2.80 \div 10 = \0.28 per pencil. $\$0.25 < \0.28 , so 6 for \$1.50 is the better buy.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).